Universidade de Aveiro Departamento de Física,
2024/2025

Sara Carlos Cardoso

The deflection of light around a black hole
A deflexão da luz em redor de um buraco negro

Universidade de Aveiro Departamento de Física,
2024/2025

Sara Carlos Cardoso

The deflection of light around a black hole
A deflexão da luz em redor de um buraco negro

Apresentado à Universidade de Aveiro para cumprimento dos requisitos necessários à obtenção da unidade curricular Projeto. Realizado sob a orientação da Drª Merce Guerrero, Investigadora Doutorada Nível 1 do Departamento de Matemática da Universidade de Aveiro.

o júri / the jury

presidente / president	**Dr. Manuel António dos Santos Barroso** Professor Auxiliar da Universidade de Aveiro
orientadora / supervisor	**Dra. Mercè Guerrero Róman** Investigadora Doutorada (Nível 1) da Universidade de Aveiro
arguente / examiner	**Dr. Héctor Raúl Olivares Sánchez** Investigador Júnior da Universidade de Aveiro

acknowledgements

Firstly, I want to express my gratitude to my supervisor, Dr. Mercè Guerrero, for all her guidance and help throughout this year. Her advice and expertise were crucial to the development of this landing place. I also want to thank my course directors, Professor Manuel Barroso and Professor Vítor Bonifácio, for their care and guidance in this challenging field of physics. Additionally, I would like to thank my family for their unwavering support, even though they may not fully understand what I'm doing. I would also like to thank my boyfriend, João, for his support over the past years, he was a great pillar of strength and motivation. I would also like to thank my friends I made along the way and the ones that were already here before.

Abstract

This project investigates light deflection around Schwarzschild black holes through computational modelling of null geodesics. Building on the principles of General Relativity, the research develops a ray-tracing framework to analyse how the curved spacetime geometry influences photon trajectories. Numerical integration of the geodesic equations reveals three distinct classes of photon behaviour. The simulations confirm this parameter as the boundary between photon capture and escape, essential for understanding the formation of black hole shadows. Transfer functions mapping observed emission to its origin in the accretion disk are calculated, demonstrating the complex gravitational lensing effects that create multiple images of the same regions. These computational results provide insights into phenomena recently observed by the Event Horizon Telescope and establish a foundation for investigating more complex black hole geometries.

Resumo

Este projeto investiga a deflexão da luz em redor de buracos negros de Schwarzschild através da modelação computacional de geodésicas nulas. Com base nos princípios da Relatividade Geral, a investigação desenvolve um enquadramento de ray-tracing para analisar como a geometria do espaço-tempo curvo influencia as trajetórias dos fotões. A integração numérica das equações geodésicas revela três classes distintas de comportamento dos fotões. As simulações confirmam este parâmetro como a fronteira entre a captura e o escape de fotões, essencial para compreender a formação de sombras de buracos negros. São calculadas funções de transferência que mapeiam a emissão observada até à sua origem no disco de acreção, demonstrando os complexos efeitos de lente gravitacional que criam múltiplas imagens das mesmas regiões. Estes resultados computacionais fornecem perspetivas sobre fenómenos recentemente observados pelo Event Horizon Telescope e estabelecem uma base para investigar geometrias mais complexas de buracos negros.

Contents

Chapter 1

Introduction

The study of light behaviour in curved spacetime represents one of the pillars of modern astrophysics, studying the nature of gravity and the structure of our universe. This research project investigates how light deflects around a black hole, with a particular focus on the formation of black hole shadows through computational modelling of photon trajectories in Schwarzschild spacetime. Since Albert Einstein published his theory of General Relativity in 1915, the understanding of gravity has undergone a fundamental shift from Newton's concept of a force acting at a distance to a geometric phenomenon arising from the curvature of spacetime. Einstein's revolutionary framework established that massive objects deform the fabric of spacetime, and this distortion determines how objects—including light—move through the universe. The Equivalence Principle asserts that acceleration and uniform gravity are indistinguishable in sufficiently small regions of spacetime, providing a foundational framework for understanding gravitational phenomena. The mathematical formulation of General Relativity centres on metrics that describe the geometry of spacetime. Among the known exact solutions to Einstein's field equations, the Schwarzschild metric holds particular significance as it describes the spacetime around non-rotating, spherically symmetric mass distributions, including black holes. This metric reveals critical features such as the event horizon and photon sphere, which profoundly influence the behaviour of light in strong gravitational fields. The theoretical predictions of GR regarding light deflection have received remarkable empirical validation. From Eddington's 1919 solar eclipse observations to the recent groundbreaking achievement of the Event Horizon Telescope (EHT) in 2019, which captured the first direct image of a black hole shadow, observational evidence continues to confirm Einstein's theory with extraordinary precision.

In addition to the theoretical part, this project aims to develop a computational framework for modelling photon trajectories in Schwarzschild spacetime using numerical integration of the geodesic equations, and to classify and characterise different types of photon behaviour based on their impact parameter. Verify the theoretical prediction of the critical impact parameter as the boundary between photon capture and escape, calculate transfer functions mapping observed emission and visualise the formation of black hole shadows and photon rings. The central hypothesis under examination is that light rays can be classified into distinct classes based on their impact parameters, with each class exhibiting characteristic behaviour that contributes differently to observable features in black hole imaging. This investigation employs ray-tracing techniques implemented through computational methods in Python. The non-linear differential equations governing null geodesics in curved spacetime are used to solve the problem using numerical methods. Ray-tracing enables the precise tracking of photon paths, both backwards and forward in time, which is essential for understanding the relationship between observed images and source regions. Computational modelling enables exploration of the entire parameter space of impact values with high precision, particularly in the critical regime near the photon sphere. Visualisation of complex trajectories provides an intuitive understanding of abstract theoretical concepts. The numerical framework utilises adaptive integration methods with high-order accuracy and tight tolerances to ensure reliable results, particularly in regions of rapidly changing behaviour near the photon sphere.

This thesis is organised into six chapters. The introduction presents the context, objectives, and methodological approach of the research. The second chapter on General Relativity provides the theoretical foundation, introducing the Equivalence Principle, curved spacetime, and Einstein's field equations that describe the relationship between matter and the geometry of spacetime. The third chapter on the Schwarzschild Solution examines the mathematical properties of the Schwarzschild metric, which represents the spacetime around a static, spherically symmetric black hole, and derives the geodesic equations for particle motion. The fourth chapter on the Ray-tracing Method develops the ray-tracing framework, reformulating the geodesic equations in terms of the impact parameter and establishing the theoretical basis for black hole shadows and photon rings. The fifth chapter, on computational methods and results, presents the numerical implementation and findings, including visualisations of photon trajectories, analysis of the critical impact parameter, and calculation of transfer functions. The final chapter summarises the key results, discusses their implications for black hole imaging, and suggests directions for future research. Through this structure, the thesis progresses from fundamental theory to computational implementation and analysis, culminating in insights about the behaviour of light in the strong gravitational field of a black hole, a phenomenon at the intersection of General Relativity, computational physics, and observational astronomy.

Chapter 2

General Relativity

The Theory of General Relativity is a revolutionary theory of gravity developed by Albert Einstein and published in 1915. Newton's understanding of gravity was based on classical field theory, where a particle of mass M would create a gravitational potential. On the other hand, Einstein proposed that gravity is understood as the deformation of spacetime resulting from the presence of matter and energy [1]. This revolutionary perspective enabled physicists to understand better gravitational effects, such as the anomalous precession of Mercury's orbit, and predict new phenomena, including the bending of light by massive objects and the existence of black holes.

General Relativity has become one of the pillars of modern physics, opening new areas of theoretical and observational research. In this chapter, we will explore the foundations of General Relativity, its mathematical formulation, and its key physical predictions. The content of this Chapter is predominantly based on the reference [1].

2.1 The Equivalence Principal

Gravity has a special property that distinguishes it from the other fundamental forces. In a gravitational field, all objects, given the same initial velocity, follow the same trajectory [2]. This fundamental observation leads to the formulation of the Equivalence Principle, which exists in three primary forms.

2.1.1 Weak Equivalence Principle

The Weak Equivalence Principle (WEP) states that the inertial mass (m_i) and the gravitational mass (m_g) of any object are equal. This implies that the trajectory of a freely falling test particle is independent of its mass and composition. The WEP states that the inertial mass and the gravitational mass of any object are equal in magnitude. This implies that the trajectory of a freely falling test particle is independent of its mass and composition.

Using Newton's Second Law of Motion and the Law of Gravitation, respectively:

$$\vec{F} = m_i\vec{a} \quad \text{and} \quad \vec{F_g} = -m_g\nabla\Phi. \tag{2.1}$$

Since experimental evidence confirms that $m_i = m_g$, it follows that all objects experience the same acceleration in a gravitational field Φ:

$$\vec{a} = -\nabla\Phi. \tag{2.2}$$

Galileo first tested this principle in his famous experiments at the Leaning Tower of Pisa and later confirmed with high precision by Eötvös-type experiments [3]. In Newtonian mechanics, WEP ensures that gravity can be described as a force acting uniformly on all bodies. However, its more profound implications extend into the framework of General Relativity.

2.1.2 Einstein Equivalence Principle

The Einstein Equivalence Principle (EEP) extends WEP by incorporating the behaviour of all non-gravitational physical laws in freely falling reference frames. It consists of three key statements. Firstly, in sufficiently small regions of spacetime, the laws of physics are reduced to those of Special Relativity. Secondly, the outcome of any local non-gravitational experiment in a freely falling frame is independent of the presence or strength of an external gravitational field. And finally, it is impossible to distinguish, utilising local experiments, between acceleration and a uniform gravitational field.

EEP implies that gravity cannot be described as a conventional force but must instead be understood as an effect of spacetime curvature. This principle forms the foundation of General Relativity, where free-falling observers experience no gravitational effects locally, analogous to inertial observers in Special Relativity.

2.1.3 Strong Equivalence Principle

The Strong Equivalence Principle (SEP) generalises the EEP by including gravitational interactions in its scope. It states that the WEP holds for all bodies, including self-gravitational systems, and that the outcome of any local experiment, whether gravitational or non-gravitational, is independent of the presence of a gravitational field. Additionally, gravity is entirely geometric and must be attributed to the curvature of spacetime.

Unlike the EEP, which applies to non-gravitational physics, the SEP asserts that the gravitational laws remain unchanged in any freely falling reference frame. This means that even complex systems, such as stars and black holes, should follow the same principles [1].

One crucial consequence of SEP is the breakdown of global inertial frames in curved spacetime. Due to inhomogeneities in gravitational fields, inertial frames can only be locally defined in small enough regions. This leads to the modern view that spacetime is a dynamic entity whose curvature dictates the motion of objects via the Einstein Field Equations. The solution is to retain the notion of an inertial frame, but such frames cannot extend throughout spacetime; they are only valid in infinitesimally small regions of spacetime.

2.2 Curved Spacetime and the Einstein Field Equations

In the previous sections, it was shown that gravity must be understood not as a force, but as a geometric property of spacetime. One of the most famous and simplest ways to explain and visualise gravity is by considering a rubber sheet, which represents spacetime. If a small mass is placed on the sheet, it warps slightly. However, if it's placed under a heavy object, the sheet deforms and creates a curved surface.

To understand the physical and geometric content of GR, it is essential first to establish the mathematical framework in which it is formulated. Throughout this section, the fundamental tools will be introduced.

2.2.1 Manifolds and the Metric Tensor in GR

General Relativity is formulated within the framework of differential geometry, where the underlying structure of spacetime is represented by a mathematical object called a manifold. A manifold is a topological space that may be curved or possess a complex global structure, but locally resembles flat Euclidean space [1]. This property allows us to apply the tools of calculus in small regions, despite the possible curvature of the overall space. A familiar analogy is the surface of the Earth, which, although globally spherical, appears flat to observers within a sufficiently small area.

In General Relativity, spacetime is modelled as a four-dimensional Lorentzian manifold, meaning it has a structure that allows for both spatial and temporal measurements (ct, x, y, z). Spacetime is represented by a pseudo-Riemannian manifold $(\mathcal{M}, g_{\mu\nu})$, where $\mathcal{M}$ is the spacetime manifold, a four-dimensional smooth surface and $g_{\mu\nu}$, the metric tensor, a symmetric (0,2) tensor. Unlike Euclidean spaces, the metric tensor $g_{\mu\nu}$ varies from point to point, allowing spacetime to be curved. The central concept on this is that the line element, or spacetime interval between two infinitesimally close events, is given by:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \tag{2.3}$$

Where $g_{\mu\nu}$ is the metric tensor, encoding information about the geometry of spacetime, and dx^μ and dx^ν represent differential coordinate displacements in four-dimensional spacetime. The metric governs not only distances and angles but also causal relationships between events, whether they are timelike, lightlike, or spacelike separated.

Crucially, the Equivalence Principle guarantees that in sufficiently small regions of spacetime, one can always choose coordinates such that the effects of curvature become negligible, and the metric locally approximates the flat Minkowski form:

$$g_{\mu\nu} \approx \eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1). \tag{2.4}$$

This local flatness means that Special Relativity holds in freely falling frames, even though the global geometry of spacetime may be curved. The main representation of the metric is

$$ds^2 = -cdt^2 + dx^2 + dy^2 + dz^2. \tag{2.5}$$

In the absence of mass or energy, the spacetime is flat, and the metric reduces globally to the Minkowski metric. However, when the presence of matter or energy curves spacetime, the metric takes a more complex form [1]. A key example is the Schwarzschild metric, which describes the geometry surrounding a static, spherically symmetric mass. This will be explored in detail in the next chapter.

2.2.2 The Einstein Field Equations

The Einstein Field Equations (EFE) form the foundation of General Relativity, linking the geometry of spacetime with its physical content. They create a system of ten interconnected and complex equations that describe how the presence of matter and energy curves spacetime. Their solution provides the metric tensor $g_{\mu\nu}$, which determines how distances and time intervals are measured in curved spacetime [2]. Mathematically, they are expressed as:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{2.6}$$

Here, $G_{\mu\nu}$ is the Einstein tensor, encapsulating the geometric properties of spacetime, and $T_{\mu\nu}$ is the stress-energy tensor, describing the energy, momentum, pressure, and stress of matter and radiation [2]. The constants G and c are Newton's gravitational constant and the speed of light, respectively. The Einstein tensor is defined as:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R. \tag{2.7}$$

The Ricci tensor $R_{\mu\nu}$ arises from contracting the Riemann curvature tensor, which measures how vectors are altered when parallel transported around infinitesimal loops. The Ricci scalar $R = g^{\mu\nu} R_{\mu\nu}$ gives a scalar measure of curvature by further contracting the Ricci tensor with the metric $g_{\mu\nu}$ [1].

The solution of the EFE provides the metric tensor $g_{\mu\nu}$, which determines how distances and time intervals are measured in curved spacetime [2]. Once the metric is known, it becomes possible to calculate the trajectories of particles and light, as well as to understand the causal structure of spacetime. That is, which events can influence others and how gravitational effects shape their motion. One can interpret the left-hand side of the EFE as describing the geometry of spacetime and the right-hand side as describing its energy-momentum content. Thus, the EFE embody a profound unity between geometry and physics.

In the special case of vacuum, i.e., in the absence of matter and energy, the stress-energy tensor vanishes:

$$T_{\mu\nu} = 0 \quad \Rightarrow \quad G_{\mu\nu} = 0. \tag{2.8}$$

This simplification plays a crucial role in deriving exact solutions such as the Schwarzschild metric.

2.2.3 The Relationship Between Matter and Spacetime Curvature

The Einstein Field Equations (2.6) demonstrate the central Principle of General Relativity so that gravity is not a force (2.1) but rather a manifestation of the curvature of spacetime caused by the presence of mass and energy [2]. This shift in perspective is encoded in the Einstein Field Equations [2]. In this framework, the apparent gravitational pull of a massive object is the natural motion of objects following curved paths, called geodesics, in a curved geometry. Unlike Newtonian gravity, general relativity describes gravity purely geometrically, mass and energy curve spacetime, and this curvature determines how objects move [2]. Particles and light travel along geodesic paths that locally extremise the spacetime interval. These geodesics are determined entirely by the spacetime metric, meaning that a particle in free fall is not being "pulled" but is simply moving along the straightest possible path available in the curved geometry [2]. In this sense, gravity is no longer a force but the result of living in a curved spacetime.

Physicist John Archibald Wheeler elegantly summarises this profound idea:

> "Matter tells spacetime how to curve, and curved spacetime tells matter how to move."
>
> — John Archibald Wheeler

To see this idea in action, the following chapters examine a specific and physically significant solution of the Einstein Field Equations, the Schwarzschild metric. This solution describes the spacetime geometry outside a static, spherically symmetric, non-rotating, and uncharged mass [2]. It provides the foundation for analysing geodesic motion, gravitational time dilation, and light-bending phenomena that, while once attributed to gravitational force, now emerge naturally from spacetime geometry.

Chapter 3

The Schwarzschild Solution

3.1 The Schwarzschild Metric

The content of this section is primarily based on the reference [2]. The Schwarzschild Solution, obtained by Karl Schwarzschild in 1916, shortly after Einstein published the Theory of General Relativity, represents the first exact solution to Einstein's Field Equations. It describes the gravitational field in the vacuum region surrounding a static, spherically symmetric, non-rotating, and uncharged massive body. The solution applies to idealised objects such as planets, stars, and black holes in the absence of external forces or intrinsic angular momentum. Under these assumptions, Einstein's vacuum field equations, where $R_{\mu\nu} = 0$, admit the following solution, known as the Schwarzschild metric:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2. \tag{3.1}$$

In this expression, G is the gravitational constant, M is the mass of the central object, c is the speed of light and (t, r, θ, ϕ) are the Schwarzschild coordinates. To simplify the expressions, the simplified units are used in which $G = c = 1$, reducing the metric to:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta, d\phi^2. \tag{3.2}$$

This form of the metric provides the foundation for analysing the structure of spacetime around a Schwarzschild Black Hole.

3.1.1 Properties of the Schwarzschild Metric

The Schwarzschild metric exhibits several vital properties that reveal the structure of spacetime around a spherically symmetric, non-rotating mass.

In the limit when $r \to \infty$, the Schwarzschild metric reduces to the Minkowski form, corresponding to flat spacetime. This asymptotic behaviour is consistent with the expectation that spacetime becomes flat far from a localised mass distribution.

The quantity $r_s = \frac{2GM}{c^2}$ defines the Schwarzschild radius, which marks the location of the event horizon, a null hypersurface that acts as a one-way causal boundary. At $r = r_s$, the metric components appear to diverge, suggesting a singularity. However, this divergence is not physical. For physical observers, the event horizon marks the limit beyond which nothing escapes. From the perspective of an external observer, any infalling object appears to only approach the event horizon without ever crossing it. However, a falling observer would reach the horizon in finite proper time, not noticing any anomalies in the surface. The apparent singularity is a coordinate singularity that arises from the choice of Schwarzschild coordinates.

In contrast, the point $r = 0$ corresponds to a true curvature singularity. This singularity is a hallmark of classical general relativity, signifying the limits of the theory in describing the internal structure of a black hole. It is unknown what happens if an observer passes the singularity at $r = 2M$, and even less so if they reach $r = 0$, but there are several interesting theories.

3.2 Geodesic Equation on the Schwarzschild Metric

In General Relativity, the motion of free particles, light rays, and others is described by geodesics. These are the paths that extremise the spacetime interval, behaving like a straight line in the curved geometry. In the case of the Schwarzschild metric, the geodesic equation provides critical insight into the behaviour of objects under the gravitational field alone. The general form of the geodesic equation is given by:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\tau}\frac{dx^\rho}{d\tau} = 0, \tag{3.3}$$

where $\Gamma^\mu_{\nu\rho}$ are the Christoffel symbols computed from the metric, x^μ are the coordinates, and τ is the affine parameter for massless particles, such as photons. The derived quantities using the Euler-Lagrange formalism from the Lagrangian:

$$\mathcal{L} = -\left(1 - \frac{2M}{r}\right)\dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2, \tag{3.4}$$

where dots denote derivatives to proper time τ. Due to the symmetries of the Schwarzschild metric, the Lagrangian admits two conserved quantities, the energy E and the angular momentum L of the particle. These are given by:

$$E = \left(1 - \frac{2M}{r}\right)\dot{t} \quad \text{and} \quad L = r^2\dot{\phi}. \tag{3.5}$$

These conserved quantities will play a central role in defining the impact parameter, which will be discussed in detail later.

Substituting E and L into the original Lagrangian allows us to express it in terms of the radial coordinate and the effective potential:

$$\mathcal{L} = \left(1 - \frac{2M}{r}\right)^{-1}\left(\dot{r}^2 - E^2\right) + \frac{L^2}{r^2}, \tag{3.6}$$

where the freedom is used to restrict our analysis to the equatorial plane ($\theta = \pi/2$) without any loss of generality, due to the spherical symmetry of the Schwarzschild solution, this is a direct consequence of the fact that geodesic motion in a spherically symmetric spacetime is always planar. By redefining the angular coordinates, the system can be oriented such that any geodesic lies within a plane, which gives rise to black holes that tilt [4]. This simplification significantly reduces computational complexity, allowing us to describe the motion using only the radial and azimuthal coordinates, r and ϕ.

The reformulation in Eq. (3.6) is particularly useful when analysing geodesic motion, as it isolates the radial dynamics and naturally introduces the concept of an effective potential [2].

3.2.1 Effective Potential and Geodesic Types

The effective potential for the Schwarzschild spacetime is given by:

$$V_{\text{eff}}(r) = \left(1 - \frac{2M}{r}\right)\left(-\delta + \frac{L^2}{r^2}\right), \tag{3.7}$$

where the constant δ is

$$\delta = \begin{cases} -1, & \text{for massive particles} \\ 0, & \text{for massless particles.} \end{cases} \tag{3.8}$$

Once the potential, depending on the type of particles, is separated, there is room to discuss different geodesics. The timelike geodesics embody the causal structure of spacetime and connect events that maintain causal relationships, ensuring that physical influences propagate following the principles of relativity [1]. Timelike geodesics lie strictly within the causal cone, ensuring that any observer following such a trajectory can maintain consistent physical interactions with their environment. Massive particles move along timelike geodesics, and those curves are the worldlines of possible observers [2]. That is, an object with mass at a stationary point. In GR, any object with non-zero rest mass moves along a timelike geodesic when no non-gravitational forces act upon it. These curves remain entirely within the light cone at each point in spacetime.

Another option to calculate the geodesics is by considering that along a path, the spacetime interval of a timelike geodesic ds^2 is negative, and the proper time, τ, measured by a clock moving along the worldline is defined as $d\tau^2 = -ds^2$ [1]. The particle's four-velocity $u^\mu = dx^\mu d\tau$ satisfies the normalisation condition:

$$g_{\mu\nu}u^\mu u^\nu = -1. \tag{3.9}$$

This condition ensures that the motion is physically realisable by a massive particle. In contrast to null geodesics, followed by photons, or spacelike curves, which have no physical particle interpretation, timelike geodesics are the only trajectories that observers or physical bodies can experience.

In astrophysical scenarios, timelike geodesics are very relevant for describing the motions of matter in an accretion disk around black holes or other compact objects. Timelike geodesics are governed by the geodesic equation (3.3). However, the study of accretion disks is out of the scope of this project. Null geodesics describe the trajectories of photons; therefore, they will be further discussed in the following chapter.

Chapter 4

Ray-tracing method

One of the most effective tools for understanding the study of photon trajectories around black holes is the ray-tracing method, which enables the reconstruction of how light propagates through curved spacetime and reaches a distant observer. Ray-tracing refers to the process of integrating the geodesic equations backwards in time from the observer's point of view, tracking the path of the photons. These trajectories follow the spacetime curvature and exhibit different behaviours, such as being directly absorbed by the black hole or orbiting multiple times before escaping. In simpler terms, the outcome of the photons is determined by their impact parameter b, a quantity that relates their angular momentum to their energy.

In this chapter, the geodesic equation is first reformulated in terms of the impact parameter and then again to implement it in the simulation. This form is then used to explore how different values of b determine the photon's fate and how this information is utilised in ray-tracing simulations to create synthetic images of black holes and other compact objects.

4.1 Geodesic Equation in terms of the Impact Parameter

The impact parameter b is defined as the quotient of the angular momentum L to the energy E of the particle:

$$b = \frac{L}{E}. \tag{4.1}$$

Physically, the impact parameter represents the distance between the direction of the incoming particle (dashed vertical line) and the centre of attraction [2], as it is shown in Fig. (4.1).

In the context of black hole imaging or gravitational lensing, the impact parameter characterises how close a photon comes to the black hole and determines whether it gets deflected or is captured [5]. Whether a photon escapes, gets deflected, or falls into a black hole is determined by the critical impact parameter $b_c = 3\sqrt{3}M$, which corresponds to the radius of the photon sphere [2]. Photons with this parameter are trapped in unstable circular orbits, forming the boundary of the black hole shadow observed in images such as those from the Event Horizon Telescope [6]. For massless particles such as photons, the spacetime interval (3.2) is null, $ds^2 = 0$. Restricting the motion to the equatorial plane ($\theta = \pi/2$), the Schwarzschild metric simplifies to, also called the null-geodesics:

$$-\left(1 - \frac{2M}{r}\right)\dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\phi}^2 = 0, \tag{4.2}$$

where overdots represent derivatives concerning an affine parameter λ [2]. Due to the symmetries of the Schwarzschild metric, energy and angular momentum are conserved, as previously

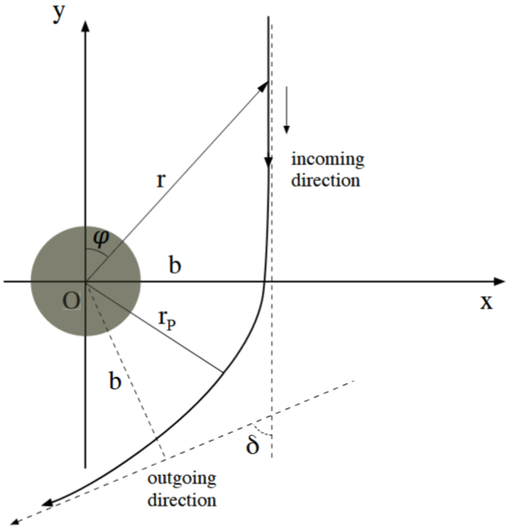

Figure 4.1: The trajectory of a massless particle deflected by a massive body.

mentioned in (3.5). Substituting these expressions into (4.2), and dividing through by E^2:

$$-1 + \frac{1}{E^2}\dot{r}^2 + \frac{b^2}{r^2}\left(1 - \frac{2M}{r}\right) = 0, \tag{4.3}$$

which can be rearranged as:

$$\dot{r}^2 = 1 - \frac{b^2}{r^2}\left(1 - \frac{2M}{r}\right). \tag{4.4}$$

To express this in terms of $\frac{dr}{d\phi}$, and applying the chain rule:

$$\dot{r} = \frac{dr}{d\lambda} = \frac{dr}{d\phi}\cdot\frac{d\phi}{d\lambda} = \frac{dr}{d\phi}\cdot\dot{\phi} = \frac{dr}{d\phi}\cdot\frac{L}{r^2}. \tag{4.5}$$

Substituting this into the expression for $\dot{r}^2$ gives:

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{r^4}{b^2}\left[1 - \frac{b^2}{r^2}\left(1 - \frac{2M}{r}\right)\right]. \tag{4.6}$$

This equation is the geodesic equation written in terms of the impact parameter. This formulation is beneficial for classifying photon trajectories and numerically integrating the path of light rays in the vicinity of black holes [7].

To further simplify the numerical implementation (A.1), it is useful to express this geodesic equation in terms of the effective potential:

$$\left(\frac{dr}{d\phi}\right)^2 = r^4\left[\frac{1}{b^2} - \frac{1 - r_s/r}{r^2}\right], \tag{4.7}$$

Here, the first term $\left(\frac{1}{b^2}\right)$ represents the centrifugal contribution, while the second term incorporates the gravitational influence of the black hole. By inverting the above equation:

$$\frac{d\phi}{dr} = \pm\frac{1}{r^2\sqrt{\frac{1}{b^2} - \frac{1-r_s/r}{r^2}}}, \tag{4.8}$$

which is the fundamental expression used in ray-tracing simulations [7]. It enables the computation of photon trajectories from the observer's emission point. It plays a crucial role in determining transfer functions, shadow boundaries, and the photon ring structure in black hole imaging.

4.2 Shadows in a Schwarzschild Black Hole

Firstly, to determine the shadows in a Schwarzschild Black Hole, there is a need to introduce a concept already mentioned in (4.6), $b_c = 3\sqrt{3}M$. But to determine this value, there is a need to look for unstable circular orbits, where:

$$\frac{dr}{d\lambda} = 0 \quad \text{and} \quad \frac{d^2r}{d\lambda^2} = 0. \tag{4.9}$$

These conditions imply that the effective potential (3.7) satisfies:

$$V_{\text{eff}}(r_c) = E^2 \quad \text{and} \quad \left.\frac{dV_{\text{eff}}}{dr}\right|_{r_c} = 0. \tag{4.10}$$

From the form of V_{eff}, we compute its derivative:

$$\frac{dV_{\text{eff}}}{dr} = \frac{L^2}{r^3}\left(\frac{6M}{r} - 2\right). \tag{4.11}$$

Setting this equal to zero gives:

$$\frac{6M}{r} - 2 = 0 \quad \Rightarrow \quad r = 3M. \tag{4.12}$$

This is the radius of the photon sphere in Schwarzschild spacetime. Substituting $r = 3M$ into the expression for $V_{\text{eff}}(r) = E^2$, we solve for the critical impact parameter:

$$E^2 = \left(1 - \frac{2M}{3M}\right)\frac{L^2}{(3M)^2} = \left(\frac{1}{3}\right)\frac{L^2}{9M^2}, \tag{4.13}$$

$$\frac{E^2}{L^2} = \frac{1}{27M^2} \quad \Rightarrow \quad \left(\frac{L}{E}\right)^2 = 27M^2 \quad \Rightarrow \quad b_c = \frac{L}{E} = 3\sqrt{3}M. \tag{4.14}$$

This result defines the critical impact parameter b_c [2]. Any photon with $b < b_c$ is captured by the black hole, while photons with $b > b_c$ are deflected and escape to infinity. Light rays arriving at the observer from very close to $b = b_c$ contribute to the bright photon ring at the edge of the black hole shadow [5].

The concept of a black hole shadow has gained importance due to recent observations by the EHT [6], which achieved an angular resolution comparable to the expected size of the massive black hole in M87. These observations sparked interest in the theoretical interpretation of the observed features, particularly the so-called black hole shadow and the surrounding photon ring [5]. In Schwarzschild's spacetime, the black hole shadow is defined by the set of directions in the sky from which no light can reach a distant observer because the light rays are captured by the black hole [5]. This boundary is traced out by light rays that asymptotically approach unstable circular photon orbits at a radius of $r = 3M$. So the corresponding critical impact parameter, which defines the apparent size of the shadow on the observer's image plane, is:

$$b_c = 3\sqrt{3}M \approx 5.1962M. \tag{4.15}$$

This curve is called the critical curve, also referred to as the apparent boundary [2]. It separates the capture photon trajectories from the ones that reach the observer. Light rays that arrive near the boundary have orbited the Black Hole due to the strong gravitational lensing [5], which contributes to the visual appearance of a bright photon ring surrounding the shadow.

The shadow of the black hole refers to the region enclosed by this critical curve [4]. In a Schwarzschild solution, where the black hole is backlit by a distant, isotropically emitting spherical screen, the shadow appears as a perfectly dark circular region, while the surroundings are uniformly bright [8].

Overall, while the shadow and photon ring provide valuable theoretical tools, the observable features of a black hole image are determined by a complex interplay between spacetime geometry, photon trajectories, and the astrophysical properties of the emitting material. Understanding these components is essential for interpreting current and future observations of black hole environments.

Chapter 5

Computational Methods and Results

This chapter presents a comprehensive numerical investigation of photon trajectories in Schwarzschild spacetime, utilising computational techniques to explore the entire parameter space of impact parameters and their corresponding orbits by modelling null geodesics. The following analysis focuses on the critical regime near the event horizon's absorption and the photon sphere, where photons can execute multiple revolutions before either escaping to infinity or falling into the black hole.

5.1 Computational Framework

The numerical investigation utilises a Python-based computational framework, where the key function is the `Scipy` Library. Python was chosen as the programming language for this project due to its versatility and extensive ecosystem of scientific libraries. It offers a balance between simplicity and computational power, making it helpful for simulations.

The core integration method is the `scipy.integrate.solve_ivp`, which provides access to multiple adaptive integration methods. Other libraries, such as `matplotlib` and `numpy`, were indispensable for constructing the simulation, allowing for precise control over numerical solvers and tolerances, and enabling the accurate treatment of photon trajectories near the event horizon.

5.1.1 Numerical Setup and Methodology

The simulation is designed to integrate the differential equations governing the null geodesics (4.2) in terms of the impact parameter (4.6). These equations describe the trajectories of photons in Schwarzschild spacetime and are solved numerically to determine their behaviour around a black hole.

Using those quantities, the differential equation for $\phi(r)$ is integrated numerically using `solve_ivp`, and is performed in both inbound and outbound directions relative to the turning point, if it exists. Different numerical integration methods were utilised to ensure robustness and accuracy across a wide range of photon trajectories. The solvers `DOP853` and `Radau`, from the `Scipy` Library, were used depending on the type of trajectory. This will be further explored in (A.1). These particular methods offer high-order accuracy and adaptive step size control in critical regions, including the photon sphere at $r = 3M$ and near the event horizon at $r = 2M$. Tight relative and absolute tolerances were set to maintain numerical stability and accurately capture the features of the trajectories, including multiple orbits. Additionally, an event detection algorithm is implemented to identify specific angular crossings of the trajectory n, where $n = \frac{\phi}{2\pi}$, where ϕ is the azimuthal angle.

Firstly, in [5], the photon rays that reach the Black Hole are classified based on how many times they cross the equatorial plane, the y-axis, which will be relevant later to characterise the observed radius r_c at each crossing.

- Direct: $n < \frac{3}{4}$
  $\frac{b}{M} \notin (5.02, 6.17)$;
- Lensed: $\frac{3}{4} < n < \frac{5}{4}$
  $\frac{b}{M} \in (5.02, 5.19)$ *or* $(5.23, 6.17)$;
- Photon Ring: $n > \frac{5}{4}$
  $\frac{b}{M} \in (5.10, 5.23)$.

The direct rays present photons that approach the black hole and are deflected only once before escaping to infinity. The lensed rays, on the other hand, are photons that orbit one or more times before escaping. The photon ring rays are photons that orbit very close to the photon sphere multiple times, forming the photon ring observed in high-resolution black hole images [6].

But what separates the rays that escape from the ones that don't is the critical impact parameter (4.15). If a certain b from a photon ray is bigger than the b_c, the ray will deflect and exit the orbit into infinity, as rays that have the impact parameter smaller than b_c are absorbed by the black hole.

5.2 Results - Photon Trajectories and Crossings

Figure (5.1) shows a vast visualisation of the photon trajectories in Schwarzschild spacetime, computed using the numerical framework described in Section (5.1). The trajectories have a wide range of impact parameters, clearly demonstrating the three classes of photon behaviour, explained in the theoretical analysis. The black disk represents the event horizon at $r = 2M$. At the same time, the dashed red circle indicates the photon sphere at $r = 3M$, which serves as the critical boundary determining the fate of photon trajectories.

The computational results validate part of the theoretical predictions about the classification of photon trajectories. The colour-coded trajectories in Figure (5.1) correspond to different impact parameter regimes, with each class exhibiting distinct geometric characteristics:

Direct Trajectories - Blue and Black The blue trajectories represent direct photon paths with impact parameters outside the critical lensing regime ($\frac{b}{M} \notin (5.02, 6.17)$). These photons approach the black hole from infinity, experience a single gravitational deflection, and escape back to infinity without completing significant orbital motion. The deflection angle increases as the impact parameter decreases, consistent with the classical expectation that closer encounters produce more substantial gravitational lensing effects. For the most significant impact parameters shown, the deflection is minimal, approaching the weak-field limit where general relativistic effects become negligible.

In contrast, the black trajectories depict photons with minimal impact parameters, which are unable to escape and instead plunge directly into the black hole. These captured rays are

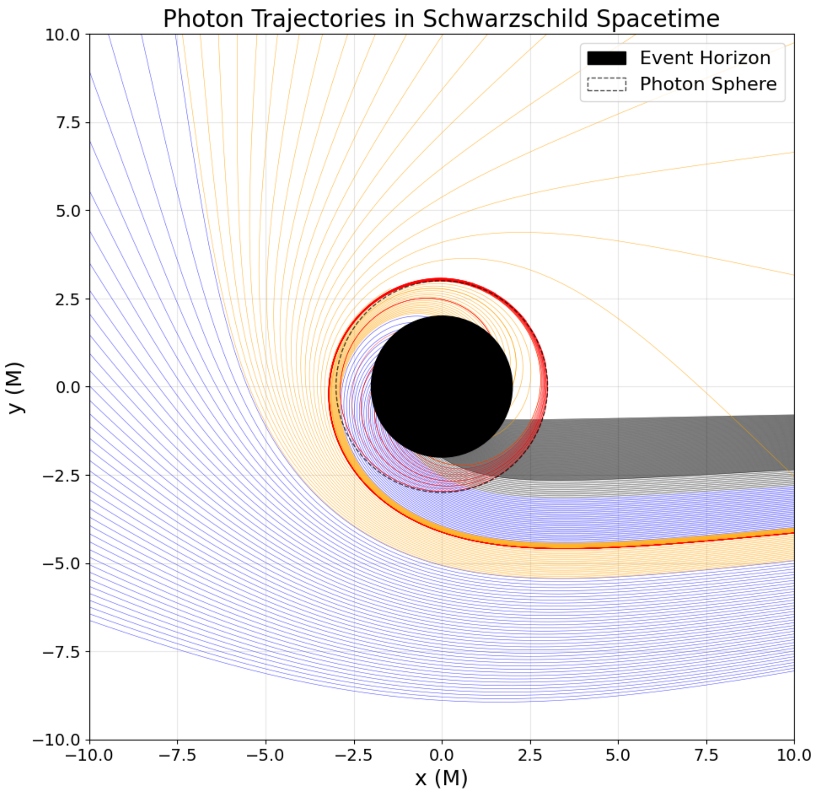

Figure 5.1: Visualisation of photon trajectories in Schwarzschild spacetime.

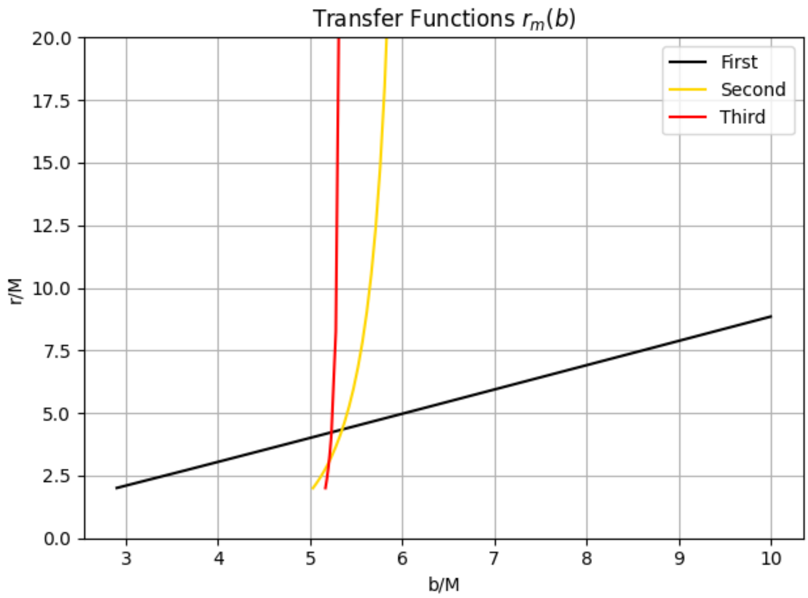

Figure 5.2: The first three transfer functions $r_c(b)$ for a face-on thin disk. Tracing a photon back from the detector, these represent the radial coordinate of the first (black), second (orange), and third (red) intersections with a face-on thin disk outside the horizon.

crucial for defining the inner shadow, a subtle but theoretically important feature of black hole imagery. While the traditional shadow is bounded by the critical curve corresponding to photon orbits, the inner shadow refers to the darker region within this boundary, formed by photons that fall into the event horizon without even crossing the equatorial plane once [4]. This feature is evident in models where the emission originates from thin equatorial disks extending to the horizon, and is visible as a central brightness depression, even within the photon ring [4].

The edge of the inner shadow coincides with the direct image of the black hole's equatorial horizon as seen through the lens. As photons in this region terminate on the horizon before sampling any of the disk's emission, they contribute no light to the observed image, resulting in an area of diminished brightness. This makes the inner shadow a unique and complementary search of the spacetime geometry near the event horizon, potentially observable in future high-dynamic-range black hole imaging efforts.

Lensed Trajectories - Orange The orange trajectories correspond to the lensed category, characterised by impact parameters in the ranges $\frac{b}{M} \in (5.02, 5.19)$ or $(5.23, 6.17)$ and angular crossings satisfying $\frac{3}{4} < n < \frac{5}{4}$. These photons execute partial orbital motion around the black hole, typically completing between one-half and one full revolution before escaping to infinity. The computational results reveal the characteristic loop structures that distinguish lensed from direct trajectories, with the loops becoming more pronounced as the impact parameter approaches the critical value.

Photon Ring Trajectories - Red The most unique features in Figure (5.1) are the tightly wound spiral trajectories shown in red, corresponding to photon ring orbits with $n > \frac{5}{4}$ and impact parameters $\frac{b}{M} \in (5.19, 5.23)$. The photons' trajectories approach the unstable circular orbit at the photon sphere, executing multiple trajectories with their radius gradually decreasing before either escaping to infinity or crossing the event horizon.

The computational results demonstrate the drastic sensitivity of photon ring trajectories to initial conditions. Subtle variations in the impact parameter within this narrow range produce dramatically different outcomes, with some photons completing dozens of orbital revolutions around the Black Hole. In contrast, others fall into the black hole after only a few circuits. This sensitivity reflects the unstable nature of the photon sphere, where trajectories exist in a delicate balance between escaping and being captured.

On the other hand, the Figure (5.2) illustrates the first three transfer functions $r_c(b)$ for a face-on thin disk. When tracing a ph ton backwards from the observer, these functions represent the radial coordinate of successive intersections, the first intersection at $\frac{-\pi}{2}$, black, the second at intersection at $\frac{-3\pi}{2}$, orange, and the third intersection at $\frac{-5\pi}{2}$, red.

These transfer functions translate the geometric relationship between observed emission and its origin in the accretion disk. The quick transitions and multiple branches visible in the functions reflect the complex gravitational lensing effects that create various images of the same disk regions, which is a phenomenon that becomes increasingly important as the observer approaches the photon sphere. The discontinuities and steep gradients in these functions mark the boundaries between different image orders, where small changes in impact parameter can alter which part of the disk contributes to the observed emission. This mathematical structure underlies the rich phenomenology of black hole imaging, from the bright photon ring to the intricate substructure revealed by high-resolution observations.

5.2.1 Critical Impact Parameter and Absorption Boundary

The numerical investigation confirms the existence of a critical impact parameter $b_c \approx 5.196M$, which serves as the boundary between trajectories that escape to infinity and those absorbed by the black hole. Photons with $b < b_c$ are captured regardless of their initial angular position, while those with $b > b_c$ eventually escape after a finite number of orbital revolutions.

The absorption boundary manifests in Figure (5.1) as the clear demarcation between trajectories that terminate at the event horizon and those that extend to the plot boundaries. The critical impact parameter lies within the photon ring regime, emphasising the fundamental role of the photon sphere in determining whether photons are captured or escape.

5.2.2 Numerical Accuracy

The computational framework exhibits remarkable numerical stability across both direct and lensed trajectories, accurately capturing the complex gravitational lensing effects with high precision. The adaptive integration methods prove essential for handling the diverse range of trajectory behaviours encountered in Schwarzschild spacetime. Unfortunately, the photon ring trajectories had a problem with the outbound trajectories, as they were not plotted. This was a problem with the outbound integration, which could not be resolved. However, it is likely due to a calculation issue. The adaptive integration methods `DOP853` and `Radau` prove particularly effective for handling the diverse range of trajectory conducts. The `DOP853` method excels for direct and lensed trajectories, where smooth evolution predominates. In contrast, `Radau` demonstrates a good performance for photon ring orbits that enter the black hole, where stiff differential equations arise near the photon sphere.

The transfer function calculations demand numerical precision, particularly in the critical impact parameter range $\frac{b}{M} \in (5.19, 5.23)$ where photon ring behaviour emerges. The resolution in (A.2) captures the rapid variations in transfer functions that create the characteristic discontinuities observed. The sharp transitions between different image orders require this enhanced sampling to resolve accurately.

Chapter 6

Conclusion

This research has explored the complex behaviour of light near a Schwarzschild black hole through computational modelling of null geodesics. Using a ray-tracing approach, it was simulated how photons travel near a non-rotating black hole and how these trajectories contribute to the optical appearance of black holes, specifically, the formation of shadows and photon rings. The numerical investigation confirmed the theoretical prediction of a critical impact parameter that separates photons that are captured from those that escape. By reformulating the geodesic equations in terms of this parameter, photon trajectories were successfully classified into three distinct categories: direct, lensed, and photon ring. Each of these plays a unique role in shaping the visual appearance of the black hole, with the photon ring forming the sharp boundary of the shadow and contributing to the bright, narrow ring seen in high-resolution images. This classification was essential for understanding the structure of the black hole shadow and the multiple-image features that arise from strong gravitational lensing.

Beyond null geodesics, the project explored the role of timelike geodesics, which are followed by massive particles, such as those in an accretion disk. While not directly simulated in this work, the mathematical groundwork was laid for future extensions. Timelike geodesics govern the orbits of particles emitting light that is then bent by the black hole's gravity, providing the source term in more complete imaging models. This makes them a natural next step in the development of the code and the theoretical framework.

The ray-tracing code developed for this project integrates geodesic equations using adaptive ODE solvers such as `DOP853` and `Radau`. This framework proved robust and accurate in simulating both direct and lensed photon paths. However, some limitations were encountered in the integration of long, outbound photon ring trajectories due to numerical instability close to the photon sphere. The transfer functions computed for multiple disk crossings revealed the intricate relationship between observed emission and its source in the equatorial plane, providing insight into the formation of photon rings observed in high-resolution black hole imaging. Overall, the simulation framework built in this project provides a solid foundation for future work in black hole imaging and gravitational lensing, with potential applications to current and upcoming data from observational initiatives such as the Event Horizon Telescope.

Future Work

Future work could extend this research in several promising directions. A natural evolution would be to incorporate more realistic astrophysical environments, particularly accretion disks with varying emission profiles and opacity distributions. Modelling the radiative transfer through these accretion structures would enable the generation of synthetic observations that more closely resemble actual black hole images, including effects such as Doppler beaming and gravitational redshift. Additionally, expanding the framework to Kerr black holes enables the study of the effects of frame dragging and the understanding of astrophysical black holes. These enhancements would bridge the gap between theoretical predictions and observational astronomy, contributing to our understanding of these extreme cosmic objects.

References

[1] Sean Carroll. *Spacetime and Geometry: An Introduction to General Relativity.* Addison-Wesley, 2004.

[2] Paolo Pani Valeria Ferrari, Leonardo Gualtieri. *General Relativity and its Applications.* CBC Press, 2020.

[3] T. Yarman, A.L. Kholmetskii, C. Marchal, O. Yarman, and M. Arik. The eötvös experiment, gtr, and differing gravitational and inertial masses proposition for a crucial test of metric theories. *Journal of Physics: Conference Series*, 1251(1):012051, jun 2019.

[4] Andrew Chael, Michael D. Johnson, and Alexandru Lupsasca. Observing the inner shadow of a black hole: A direct view of the event horizon. *The Astrophysical Journal*, 918(1):6, August 2021.

[5] Samuel E. Gralla, Daniel E. Holz, and Robert M. Wald. Black hole shadows, photon rings, and lensing rings. *Physical Review D*, 100(2), July 2019.

[6] Event Horizon Telescope Collaboration, Kazunori Akiyama, et al. First m87 event horizon telescope results. i. the shadow of the supermassive black hole. *The Astrophysical Journal Letters*, 875(1):L1, 2019.

[7] G. S. Bisnovatyi-Kogan and O. Yu. Tsupko. Strong gravitational lensing by schwarzschild black holes. *Astrophysics*, 51(1):99–111, January 2008.

[8] Samuel E. Gralla, Alexandru Lupsasca, and Daniel P. Marrone. The shape of the black hole photon ring: A precise test of strong-field general relativity. *Physical Review D*, 102(12), December 2020.

Appendix A

Python Code

This appendix highlights selected components of the Python code created to simulate the photon trajectories in Schwarzschild Spacetime. In Section (??)ection: Computational Framework, the reasons for choosing Python to simulate the numerical integration of null geodesics are already explained.

A.1 Numerical Ray-Tracing

The central structure of the code is the `SchwarzschildPhotons` Class, where the main constants are initialised.

```
class SchwarzschildPhotons:
    def __init__(self, M=1):
        self.M = M
        self.rs = 2 * M             # Schwarzschild radius
        self.r_photon = 3 * M       # Photon sphere radius
        self.b_crit = 3 * np.sqrt(3) * M   # Critical impact parameter
        ...
```

Listing A.1: Initialization of Schwarzschild geometry

These constants define the event horizon, the photon sphere, and the critical impact parameter. These definitions help classify and compare the types of trajectories later on. The geodesic equation (4.6) for a massless particle in Schwarzschild spacetime is numerically solved using `solve_ivp`. For stability and precision, it uses adaptive methods and tight tolerances.

Firstly, each photon trajectory is classified based on its impact parameter. This classification is crucial for distinguishing the various emission regimes.

```
def classify_orbit(self, b):
    if 5.02 <= b <= 5.19 or 5.23 <= b <= 6.17:
        return 'lensed'
    elif 5.190 <= b <= 5.23:
        return 'photon_ring'
    else:
        return 'direct'
```

Listing A.2: Classification by impact parameter

This segmentation allows targeted analysis and plotting of different photon populations. To determine whether a photon is absorbed or escapes by the black hole, the next function calculates the turning point using the function `brentq`, which finds a root of a function in a bracketing interval using Brent's method.

```python
def get_turning_point(self, b):
    def equation(r):
        return 1 / b**2 - (1 - self.rs / r) / r**2
    ...
    r_turn = brentq(equation, r_min, r_max)
    return r_turn if r_turn > self.rs else None
```

Listing A.3: Detection of horizon crossing

If `get_turning_point` returns `None`, the ray is deemed to have crossed the event horizon and is thus absorbed.

```python
def dphi_dr(self, r, b):
    term = (1 / b**2) - (1 / r**2) * (1 - self.rs / r)
    if term <= 0 or r <= self.rs:
        return 0
    return 1 / (r**2 * np.sqrt(term))
```

Listing A.4: Integration of photon geodesics

This function computes the derivative $\frac{d\phi}{dr}$ based on the effective potential, which has already been calculated for the null geodesics (4.6). And to integrate the inbound and outbound parts of each trajectory:

```python
def generate_trajectory(self, b, r_start=100, r_final=100):
    def ode_in(r, phi):
        return self.dphi_dr(r, b)
    sol = solve_ivp(
        ode_in,
        [r_start, r_final],
        [0.0],
        method='DOP853',
        rtol=1e-11,
        atol=1e-13,
        dense_output=True,
        max_step=0.01
    )
    return sol.t, sol.y[0]
```

Listing A.5: Integration of the photon trajectory using `solve_ivp`

and doing the same for the outbound part. The final component visualises the results in polar coordinates. Different trajectory types are plotted using distinct colours and transparency:

```python
def plot_single_trajectory(self, b, color, lw, alpha, ax):
    result = self.generate_full_trajectory(b)
    r_vals, phi_vals = result[:2]
    x_vals = r_vals * np.cos(phi_vals)
    y_vals = r_vals * np.sin(phi_vals)
    ax.plot(x_vals, y_vals, color=color, lw=lw, alpha=alpha)
```

Listing A.6: Plotting photon trajectories

This creates visual maps of photon motion, highlighting how photons with different b values interact with the Schwarzschild geometry.

A.2 Transfer functions and Crossings

The central part of this code segment is the calculation of the radial positions $r_m(b)$, at which a photon of a given impact parameter intersects the equatorial plane of a face-on thin disk in Schwarzschild spacetime. These intersections correspond to the direct image, $m = 1$, the first lensed image $m = 2$, and the photon ring $m = 3$ [5].

To compute the integration of the geodesic equation for massless particles (4.2), the changing variable $u = \frac{1}{r}$ is used. The governing equation is a second-order non-linear differential equation:

$$\frac{d^2u}{d\phi^2} + u = 3Mu^2. \tag{A.1}$$

This equation is solved numerically using `solve_ivp` over a domain in $\phi \in [0, -2\pi]$, allowing the photon to complete multiple orbits around the black hole.

```python

b_vals = np.concatenate([
    np.linspace(2.9, 5.19, 100),
    np.linspace(5.19, 5.23, 300),
    np.linspace(5.23, 10.0, 100),
])

def geodesic(phi, y):
    u, du_dphi = y
    d2u_dphi2 = 3 * M * u**2 - u
    return [du_dphi, d2u_dphi2]

target_phis = [-np.pi/2, -3*np.pi/2, -5*np.pi/2]

r1_list, r2_list, r3_list = [], [], []

for b in b_vals:
    r0 = 100
    u0 = 1 / r0
    du_dphi0 = -1 / b
    y0 = [u0, du_dphi0]

    phi_span = (0, -2 * np.pi)
    phi_eval = np.linspace(*phi_span, 10000)

    sol = solve_ivp(geodesic, phi_span, y0, t_eval=phi_eval, rtol=1e
        -9, atol=1e-9)

    phi = sol.t
    u = sol.y[0]
    r = 1 / u

    crossings = []
    for tgt_phi in target_phis:
        idx = np.argmin(np.abs(phi - tgt_phi))
        crossings.append(r[idx] if 0 <= idx < len(r) else np.nan)
    r1_list.append(crossings[0])
    r2_list.append(crossings[1])
```

```
    r3_list.append(crossings[2])

r2_list = clean_r_list(r2_list)
r3_list = clean_r_list(r3_list)

#Plot the resulting transfer functions
    ...
```

Listing A.7: Integration of null geodesics for computing disk intersections

Each curve represents the radial location of a photon's m^{th} crossing with the disk as a function of its impact parameter [2]. The first crossing corresponds to the direct image of the disk, while the second and third curves represent increasingly lensed and demagnified images, often contributing to the so-called photon ring structure in high-resolution black hole imagery.

The function `clean_r_list` ensures that only physically significant intersections, i.e., $r > 2M$ and finite, are considered. Fine sampling near the photon ring allows the code to accurately resolve the narrow range in which these higher-order images appear.